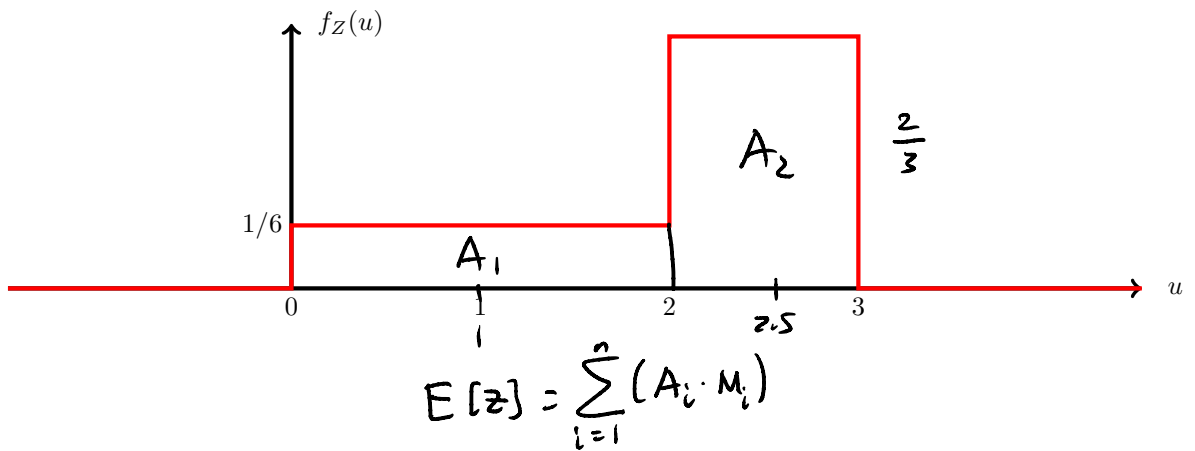


If random variable Z has probability density function shown below, then what is its expected value?



- (a) 2
- (b) 1
- (c) 3
- (d) 0
- (e) $1/2$
- (f) $5/2$
- (g) $3/2$
- (h) $3/4$
- (i) $1/3$
- (j) $6/5$
- (k) $2/3$
- (l) None of these

$$\frac{2}{6} + 1 \cdot h = 1$$

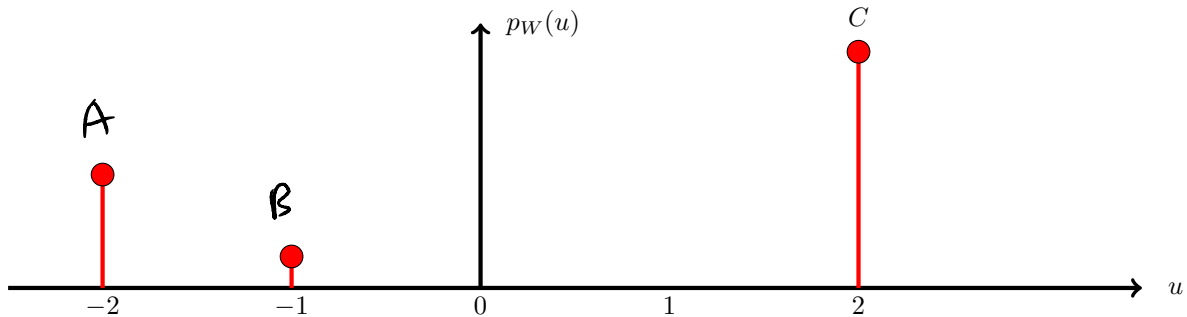
$$\rightarrow h = \frac{2}{3}$$

$$E[Z] = \frac{2}{6} + 2.5 \left(\frac{2}{3} \right)$$

$$= \frac{2}{6} + \frac{10}{6}$$

$$= 2$$

The random variable W has probability mass function shown below. Its expected value is $1/4$ and the probability that W^2 is larger than two is $11/12$. What is C ?



$$E[W] = \frac{1}{4}$$

(a) $13/24$

(b) $1/8$

(c) $3/8$

(d) $5/8$

(e) $1/24$

(f) $11/24$

(g) $5/12$

(h) $7/12$

(i) $1/6$

(j) $5/6$

(k) $3/8$

(l) None of these

$$P(W^2 > 2) = \frac{11}{12}, \quad P(W^2 \leq 2) = \frac{1}{12}$$

$$W^2 > 2 \quad @ \quad W = \{-2, 2\}$$

$$A + C = \frac{11}{12} \rightarrow B = \frac{1}{12}$$

$$E[W] = \sum u \cdot p_W(u)$$

$$E[W] = -2 \cdot A - 1 \cdot B + 2 \cdot C$$

$$\frac{1}{4} = -2A - \frac{1}{12} + 2C$$

$$\frac{1}{3} = -2A + 2C$$

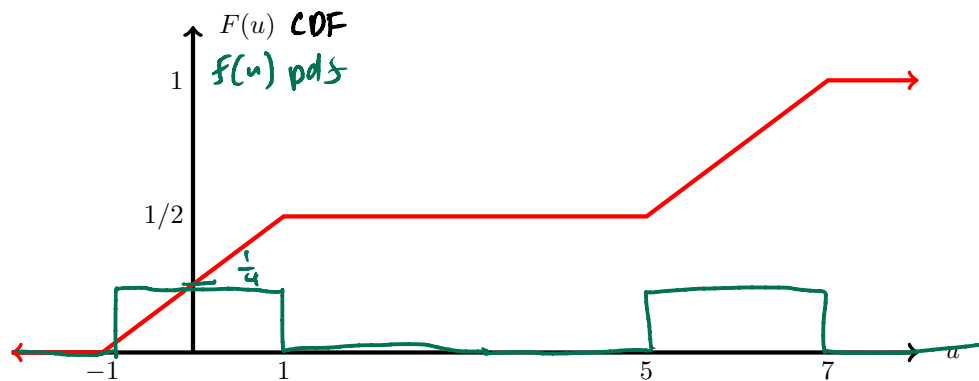
$$2A + 2C = \frac{22}{12}$$

$$+ -2A + 2C = \frac{1}{3}$$

$$4C = \frac{13}{6} \rightarrow$$

$$C = \frac{13}{24}$$

What is the variance of a random variable whose cumulative distribution function is shown below?



$$\text{Var}(U) = E[U^2] - (E[U])^2$$

$$E[U] = \sum_{i=1}^n (A_i \cdot M_i)$$

$$= \frac{1}{2}(0) + \frac{1}{2}(6) = 3$$

$$(E[U])^2 = 9$$

$$E[U^2] = \int_{-\infty}^{\infty} u^2 f_x(u) du$$

$$= \int_{-1}^1 u^2 \left(\frac{1}{4}\right) du + \int_5^7 u^2 \left(\frac{1}{6}\right) du$$

$$= \frac{1}{4} \left[\frac{1}{3} u^3 \right]_{-1}^1 + \frac{1}{6} \left[\frac{1}{3} u^3 \right]_5^7$$

$$= \frac{1}{12}(1+1) + \frac{1}{12}(7^3 - 5^3)$$

$$= \frac{1}{6} + \frac{109}{6} = \frac{55}{3}$$

$$\text{Var}(U) = \frac{55}{3} - 9$$

$$= \frac{28}{3}$$

(a) 28/3

(b) 14/3

(c) 56/3

(d) 112/3

(e) 6

(f) 3

(g) 1/3

(h) 2/3

(i) 16

(j) 8

(k) 9

(l) None of these